TIME APPLIANCES
(TAP)



A Timecard as a R-PNT custom solution

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The Timecard Paradigm shift

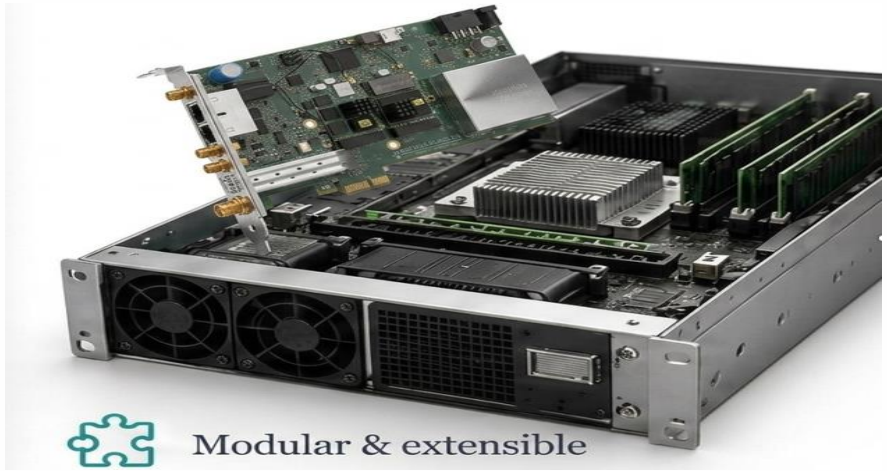
Dedicated appliance



External 1U/2U timing server

- Fixed-function hardware
- Separate rack space and power
- More cabling to servers
- Appliance-specific lifecycle

Modular timecard



PCIe module inside a COTS server

- Timing source close to compute
- Reuses server power, cooling and inventory
- Scales with open infrastructure
- Supports OCP-TAP and extended p3335 functions

Timecard Standards

TIME APPLIANCES (TAP)



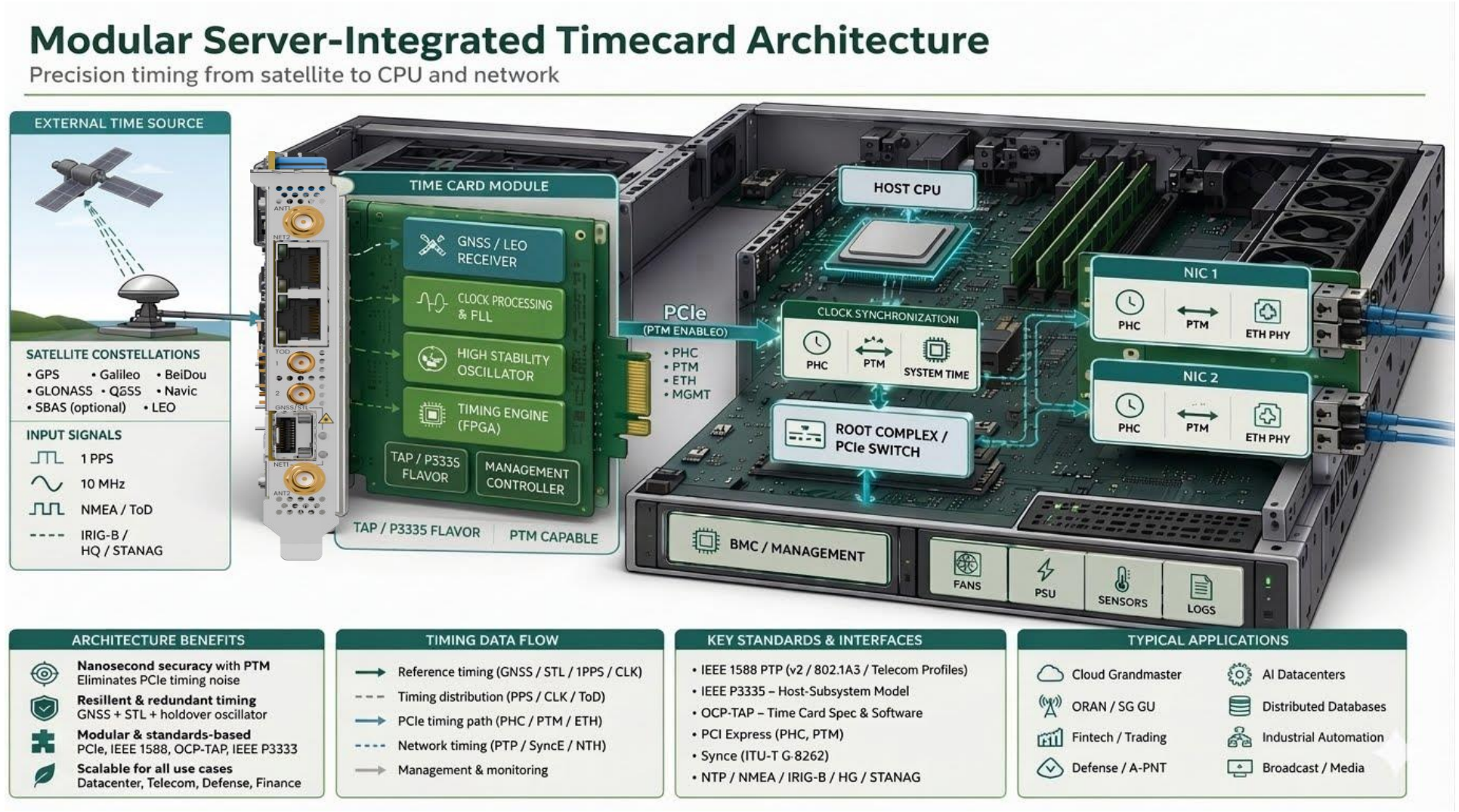
- Open specification and reference design
- Interop able implementation
- Managed by OCP-TAP driver
- Mainly functions as GNSS input to sync host system

IEEE P3335

IEEE SA STANDARDS ASSOCIATION

- Generic architecture and interfaces of a TimeCard system
- Defines optional timing inputs and outputs
- Defines optional control interfaces
- Not necessarily open architecture

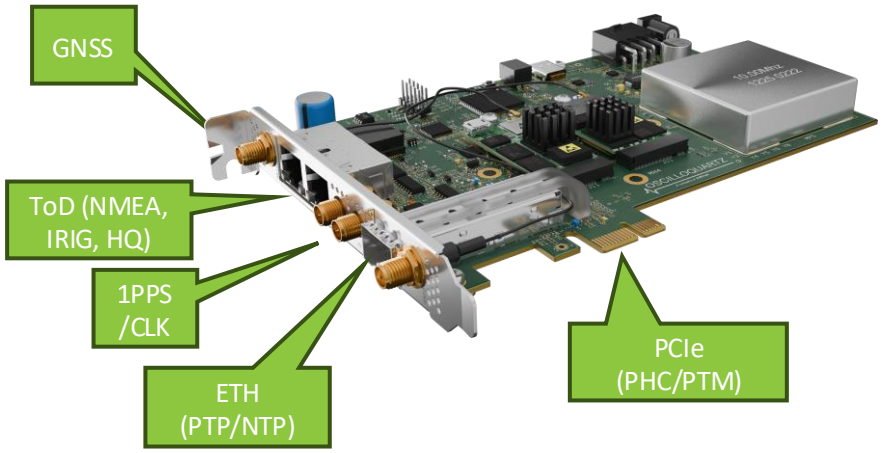
Timecard Basic Architecture



Time Card options

A timecard is not necessarily a PCIe card…

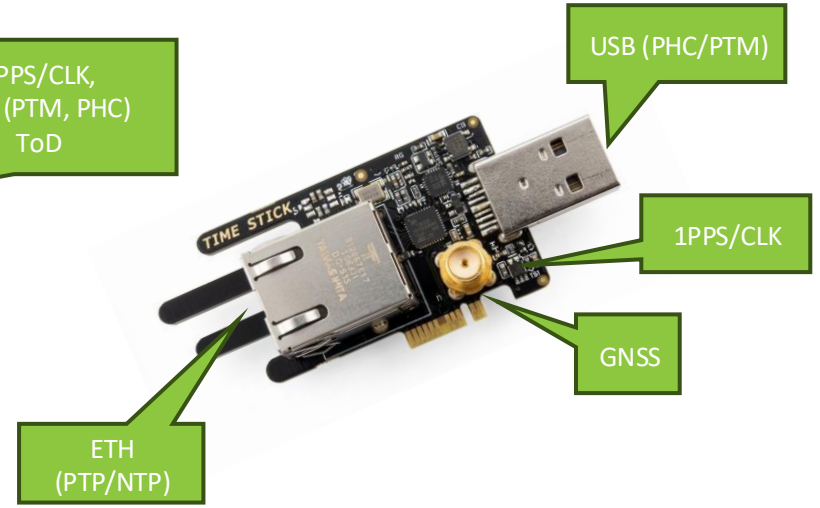
PCIe Timecard



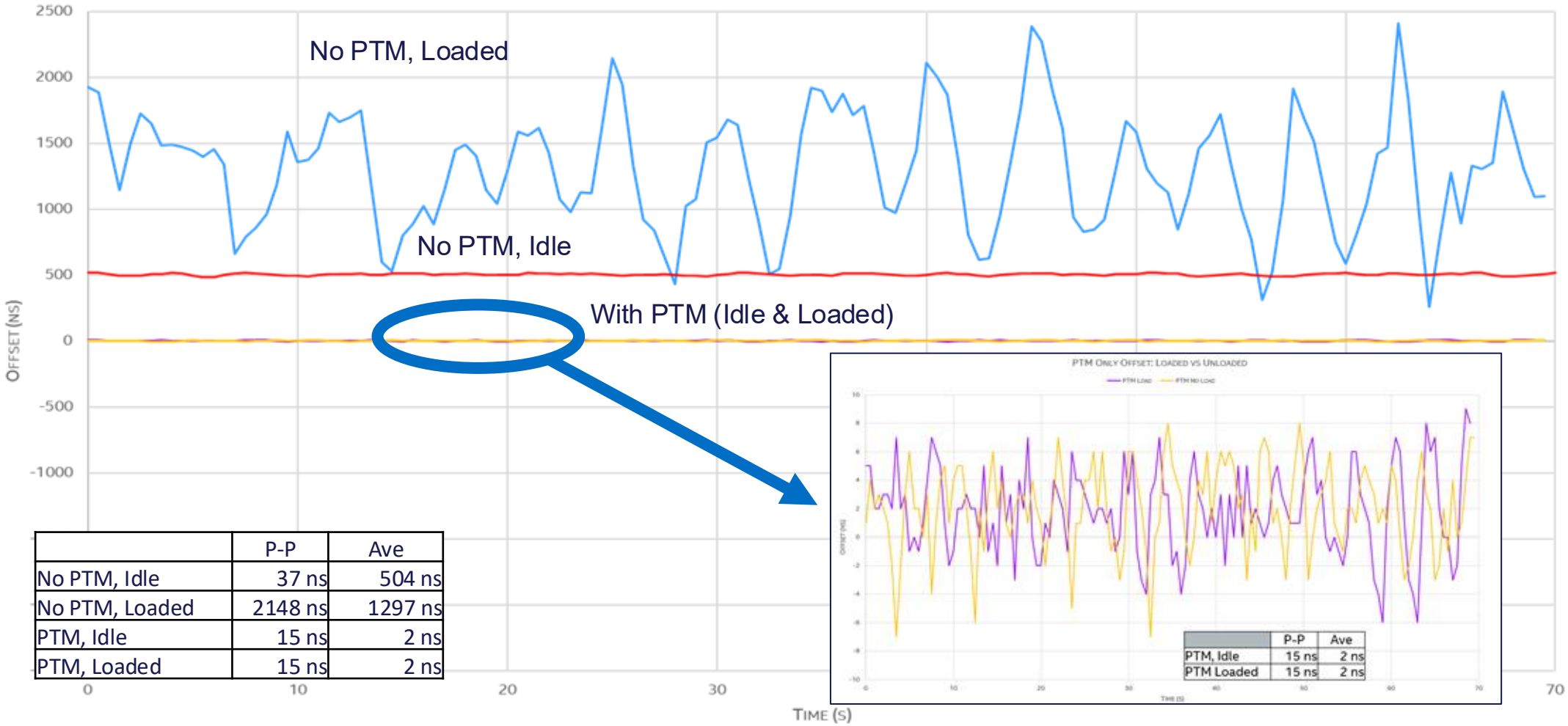
m.2 module



USB interface



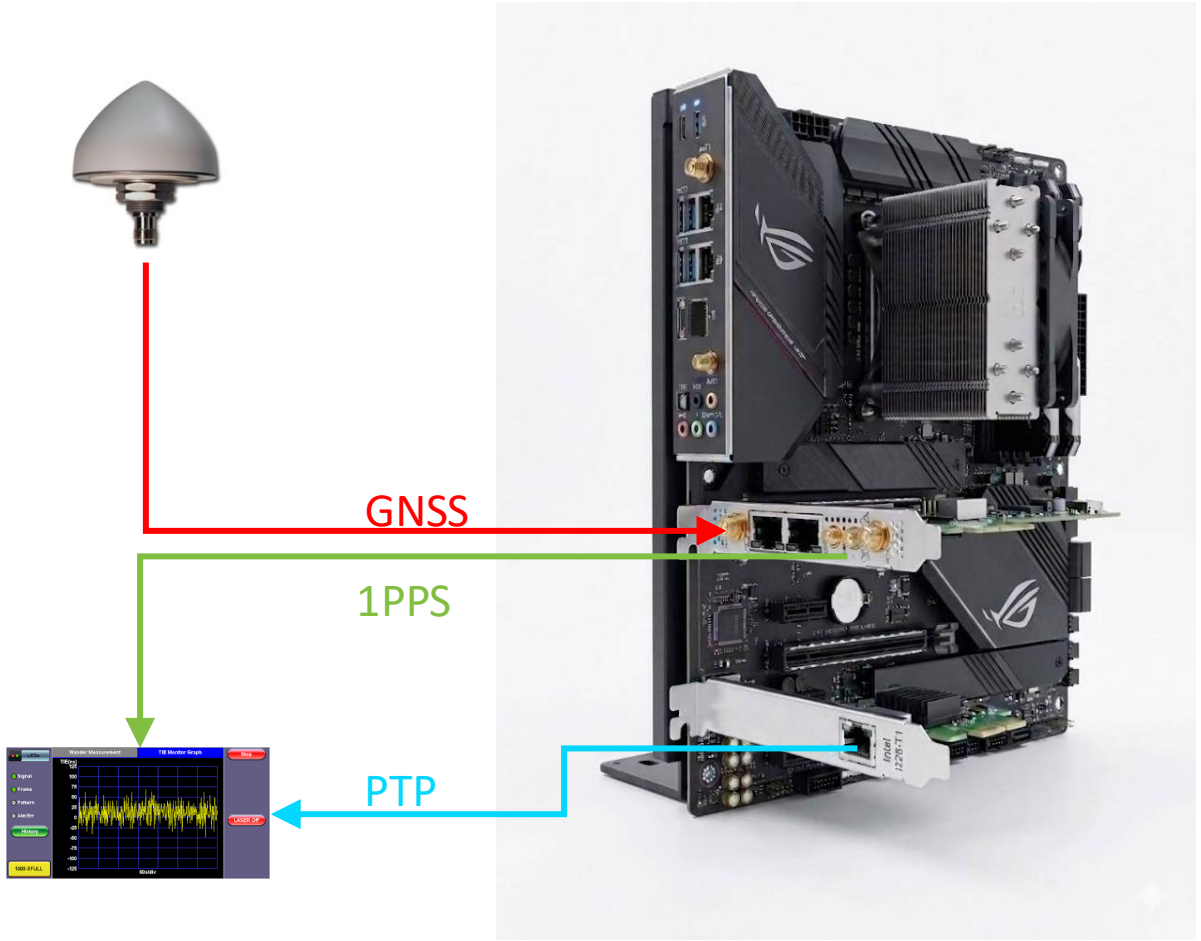
PTM (Precision Time Measurement)



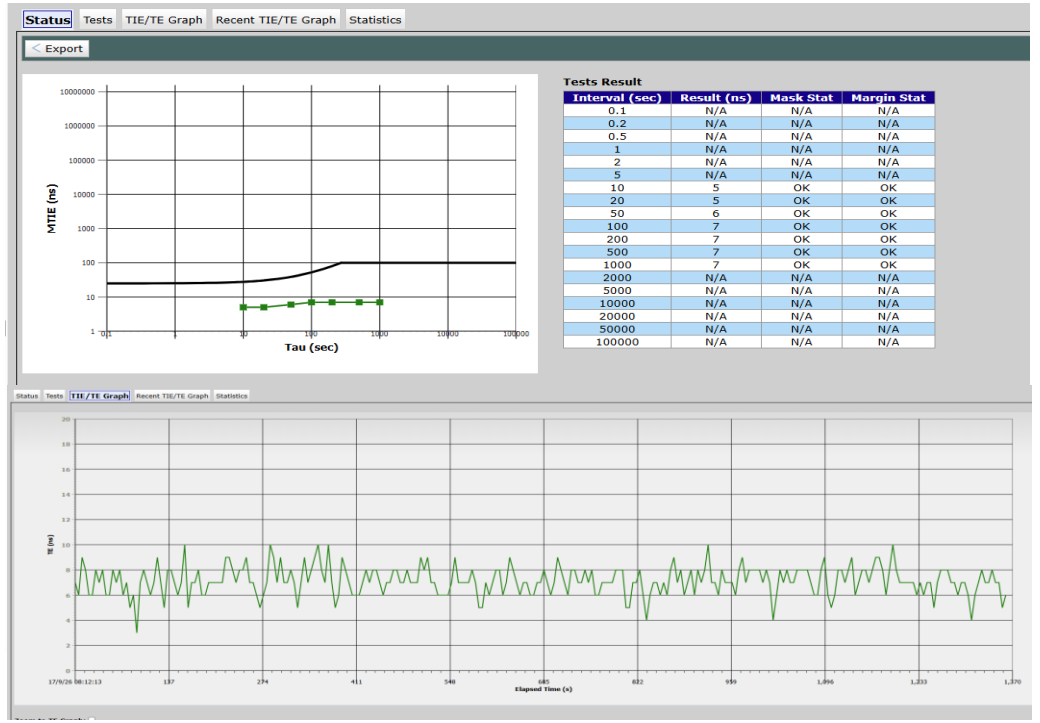
	P-P	Ave
No PTM, Idle	37 ns	504 ns
No PTM, Loaded	2148 ns	1297 ns
PTM, Idle	15 ns	2 ns
PTM, Loaded	15 ns	2 ns

From OCP TAP Webinar 2023

PRTC-B timecard system with PTM



- Timecard: OSA5400
- NIC: Intel I226
- CPU/PCH: Intel Z890
- PTM enabled on Timecard and NIC
- 1PPS: output from OSA5400 PHC
- PTP: ptp4l G.8275.1 master through I226



Timecard solutions



Data Center



Trail latency

Synchronized clocks ensure parallel GPU tasks start and end together, reducing costly wait states.



Database Throughput

Precise time enables faster transaction commits in distributed systems such as Spanner-like and CockroachDB-like architectures.



Host Synchronization

PCIe PHC2SYS and PTM synchronize the host directly for more deterministic software behavior.



Cloud Scalability

Standard servers become timing-aware nodes without deploying separate timing appliances in every rack.

Finance



Timestamp accuracy

Finance workloads demand sub-nanosecond timestamping for both competitive latency and regulatory confidence under MiFID II.

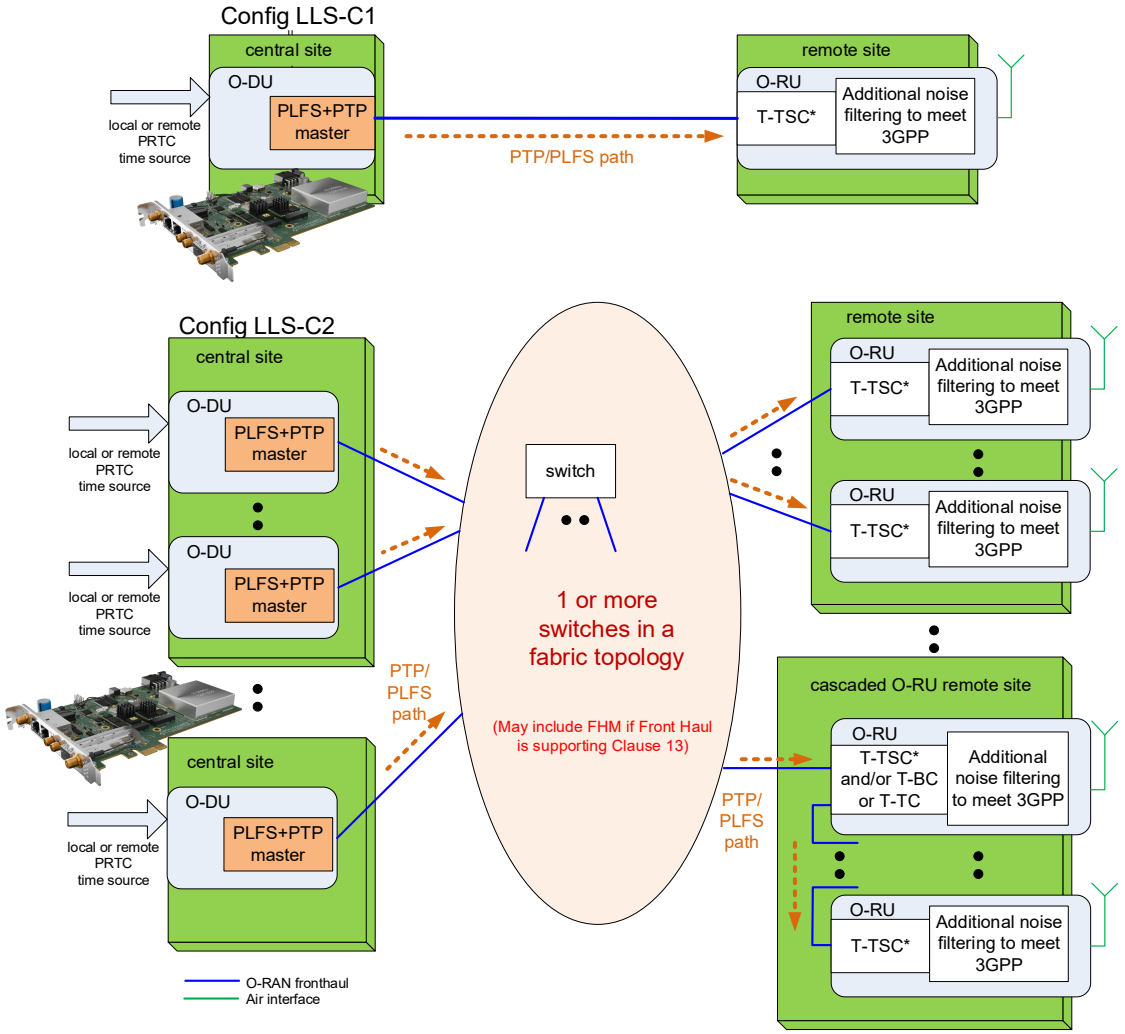
PCIe Direct

PTM eliminates microsecond-level bus jitter and delivers precise time directly to the trading algorithm.

Picosecond Ready

Timecard architecture can be designed to support White Rabbit, enabling picosecond-class timing across the trading floor.

Telecom (V-RAN, private 5G)



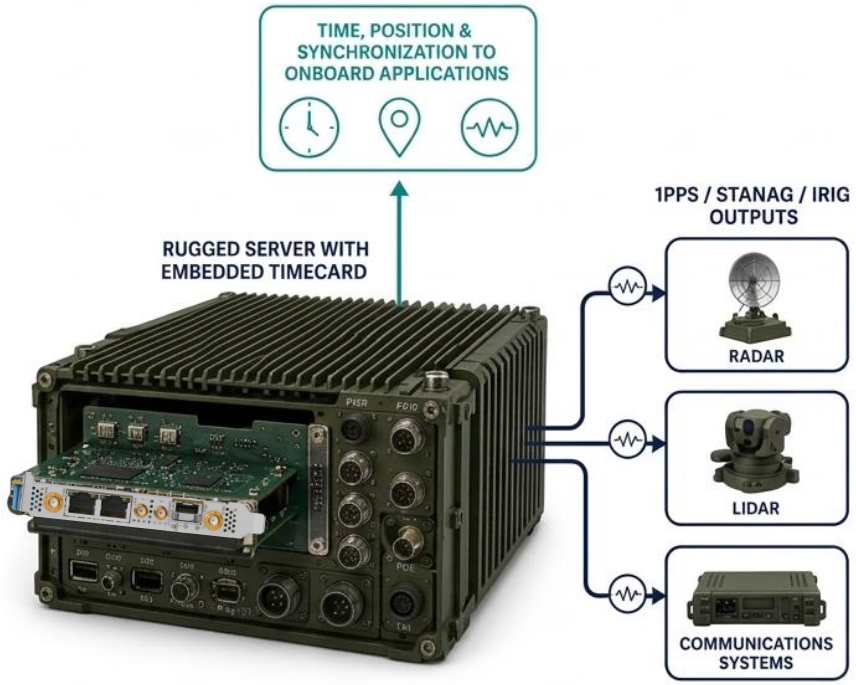
- Timecard can act a PRTC-B to the O-DU providing ns accuracy
- Timecard can act a PTP interface for O-DU (with or without NIC timestamping)
- Providing R-PNY timing solution embedded in O-DU HW/SW

Defense

Ruggedized platforms increasingly need accurate assured PNT without adding bulky standalone timing hardware.

A PCIe timecard lets an existing rugged server deliver direct timing, positioning, and external sensor synchronization.

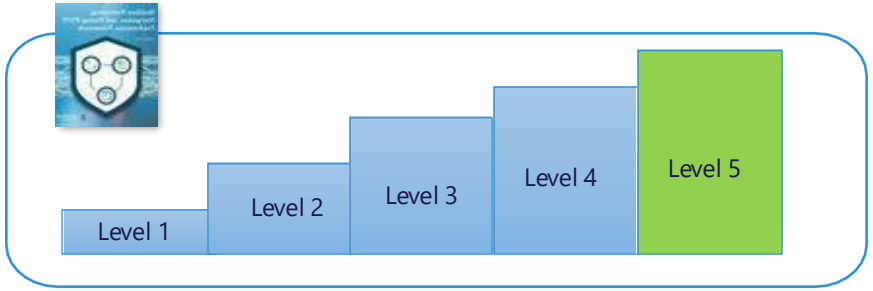
Military related timing formats (e.g STANAG, HaveQuick) can be easily supported on timecard architecture



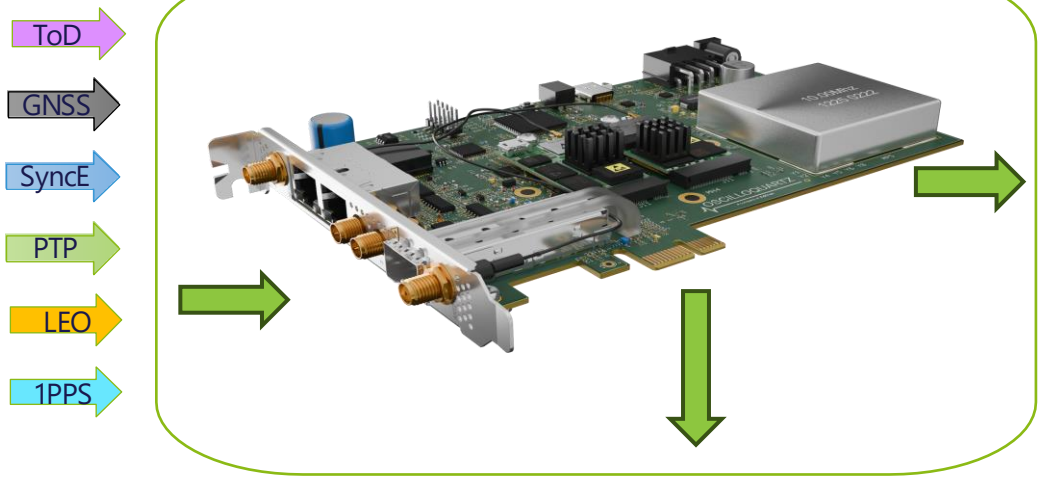
R-PNT (IEEE P1952)

Resilience Level	Requirement Summary
RL1: **Detect and Alert**	Capable of **detecting PNT Adversities** and issuing alerts that indicate the affected PNT Input Interface and meet the resilience metrics requirements for the supported Use Case.
RL2: Recover **+ Detect and Alert**	Capable of **recovery to objective criteria** within the maximum allowable recovery time interval, after detecting a PNT Adversity is no longer present. The Use Case specifies if the recovery is initiated automatically or manually.
RL3: Resist **+ Recover** **+ Detect and Alert**	Capable of meeting **minimal criteria** during a PNT Adversity, for **the required minimum time interval** for the specified Use Case.
RL4: Withstand **+ Resist** **+ Recover** **+ Detect and Alert**	Capable of **meeting objective criteria** for the specified Use Case **during a PNT Adversity**.
RL5: Verify **+ Withstand** **+ Resist** **+ Recover** **+ Detect and Alert**	Capable of **verifying the relevant PNT metrics** of the PNT Source inputs or PNT Output Solution, as defined by the specified Use Case.

A Timecard as a R-PNT solution



Multiple sources and interfaces



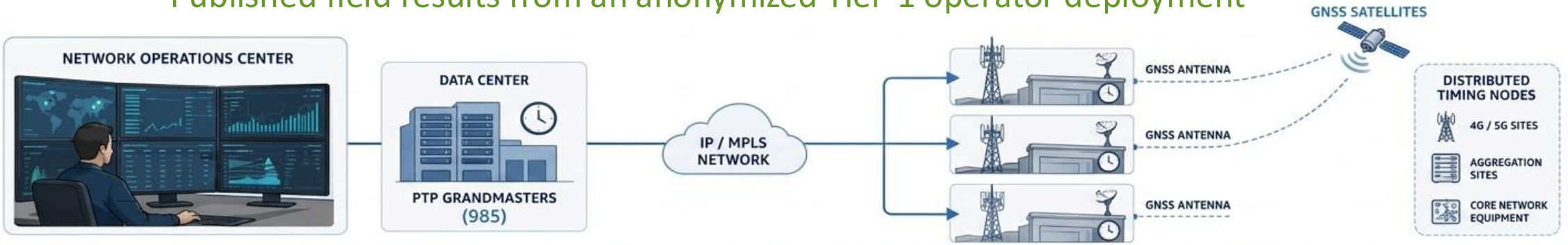
Different oscillator options for long holdover durations

Direct and local interface to host CPU provides easy integration of:

- **Multiple Source comparison and verification**
- **AI/ML routines for animalities detection (e.g. GNSS spoofing)**

Case Study: GNSS Interference in A LIVE MOBILE NETWORK

Published field results from an anonymized Tier-1 operator deployment



1 DEPLOYMENT ENVIRONMENT

- 985 PTP grandmasters
- Distributed production network
- Two monitored audit windows
- GNSS used as a primary timing reference

2 FIELD OBSERVATIONS

Alarm footprint

46.5%

46.5% of grandmasters reported alarms

458 of 985 in the later audit window

Earlier window: 20.7%

Jamming (interference)

24,022 events in two months

Monthly rate: 1,527 → 12,011



7.9× increase

89% concentrated in the top 3 affected areas

Spoofing (interference)

Average duration: 5:5 → 36.9 min/event

6.7× longer

Events/month fell 67%

3 OPERATIONAL VULNERABILITY

AVAILABILITY RISK

Frequent GNSS loss can force grandmasters into holdover or alternate-reference operation.

INTEGRITY RISK

Fewer alarms can still create prolonged exposure to an incorrect time reference.

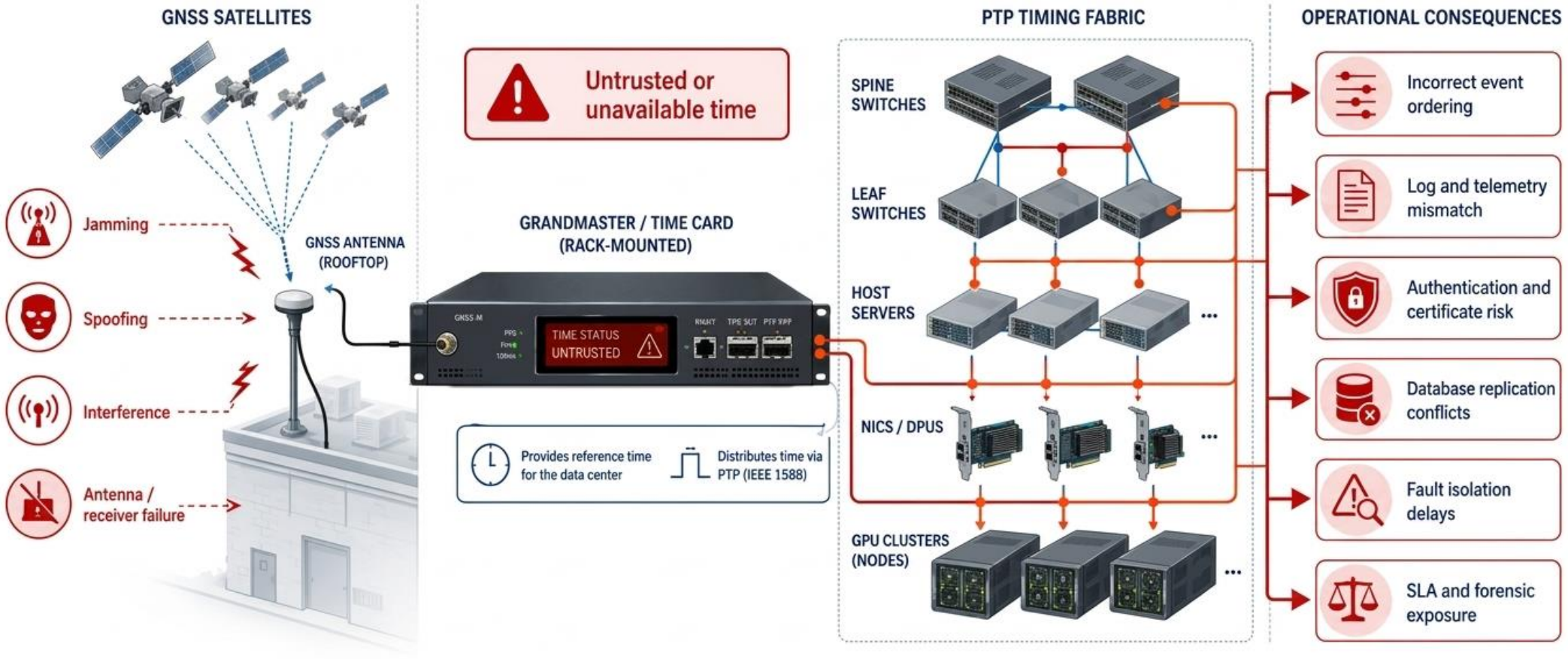
A common external GNSS dependency can affect many distributed timing nodes simultaneously.

4 INFRASTRUCTURE RESPONSE

- Detect and validate GNSS anomalies
- Use independent PTP and frequency references
- Provide high-stability holdover
- Correlate alarms across the timing network
- Quarantine untrusted time before distribution

CASE-STUDY LESSON: GNSS must be monitored as an external input—not implicitly trusted as the single source of time.

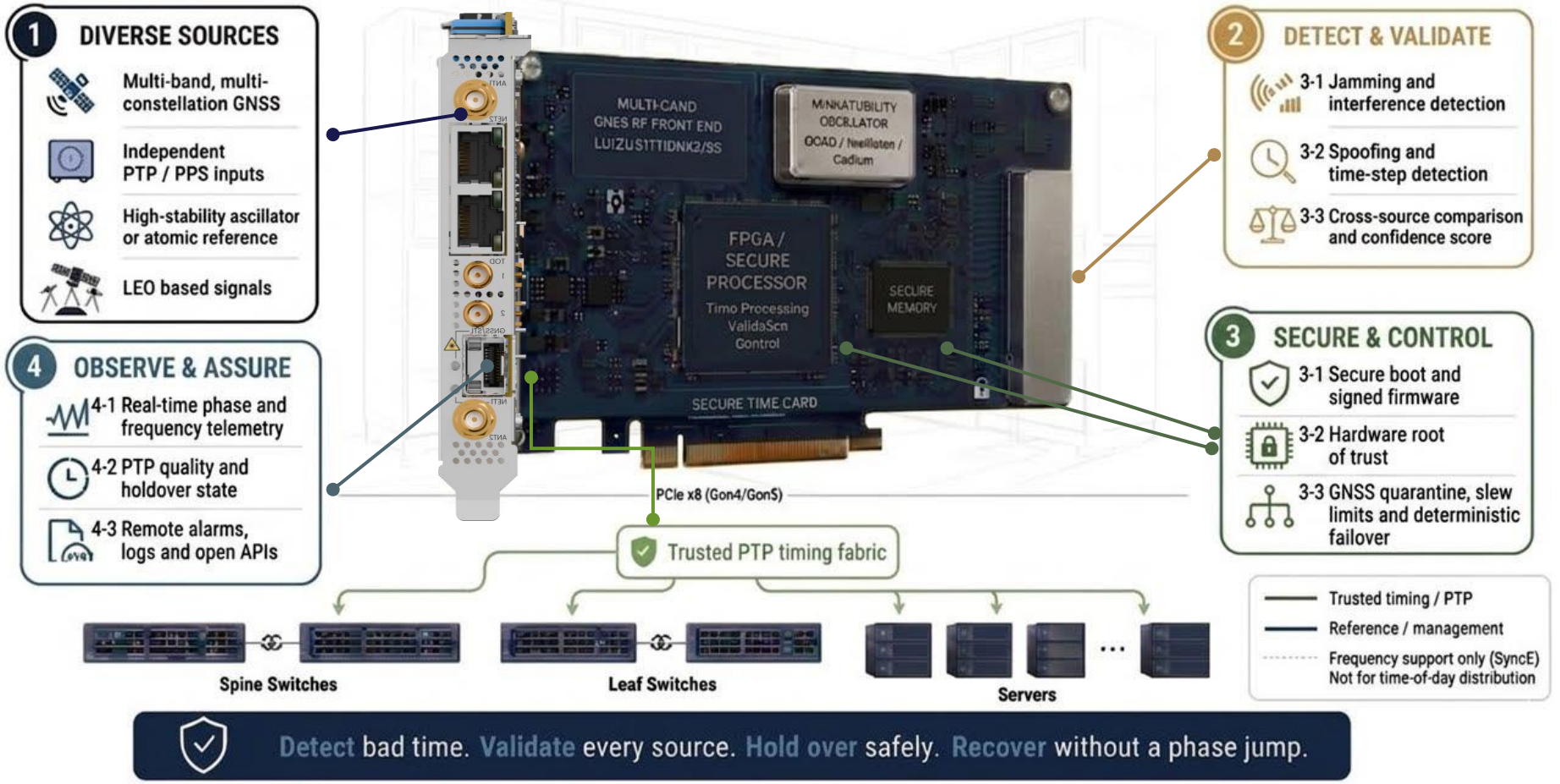
GNSS: A Shared Timing Risk in Data Centers



One vulnerable reference can become a data-center-wide failure domain.

Protected Timecard: Recommended Resilience Features

Recommended Architecture For OCP compatible PCIe Timecard



These are recommended features for resilient timing. Not part of any current OCP certification

Proposed TAP contribution: a GNSS-assured OCP TimeCard

An open assurance profile to prevent untrusted GNSS from becoming trusted data-center time.

PROBLEM TO SOLVE

Make GNSS trust explicit

- A Timecard may accept jammed, spoofed or faulty GNSS without a shared trust model.
- A common external reference can affect many servers and timing nodes simultaneously.

Community need

Standardize detection, validation, quarantine and trust-state reporting.

OSCILLOQUARTZ CONTRIBUTION

Assured profile + reference implementation

01	**Multi-source validation**	GNSS, PTP, frequency and holdover confidence
02	**Integrity controls**	Jamming/spoofing alarms and source quarantine
03	**Resilient operation**	Deterministic switching and phase-slew limits
04	**Open observability**	Telemetry, APIs, audit events and test vectors

HOW TO GET INVOLVED

1. Open the assurance use case in TAP.
2. Add operator, hyperscaler, silicon and OEM requirements.
3. Co-develop the profile, reference design and test plan.
4. Validate through TAP interoperability and OCP contribution.

Availability: contribution milestones are proposed and subject to TAP community alignment. Existing OSA 5400 Timecard is listed by the project as a compatible derivative.

CONTRIBUTION PATH

1. **Requirements** — Community alignment
2. **Specification** — Assurance profile
3. **Reference** — Firmware + telemetry
4. **Validation** — Interop + recognition

Thank You!

